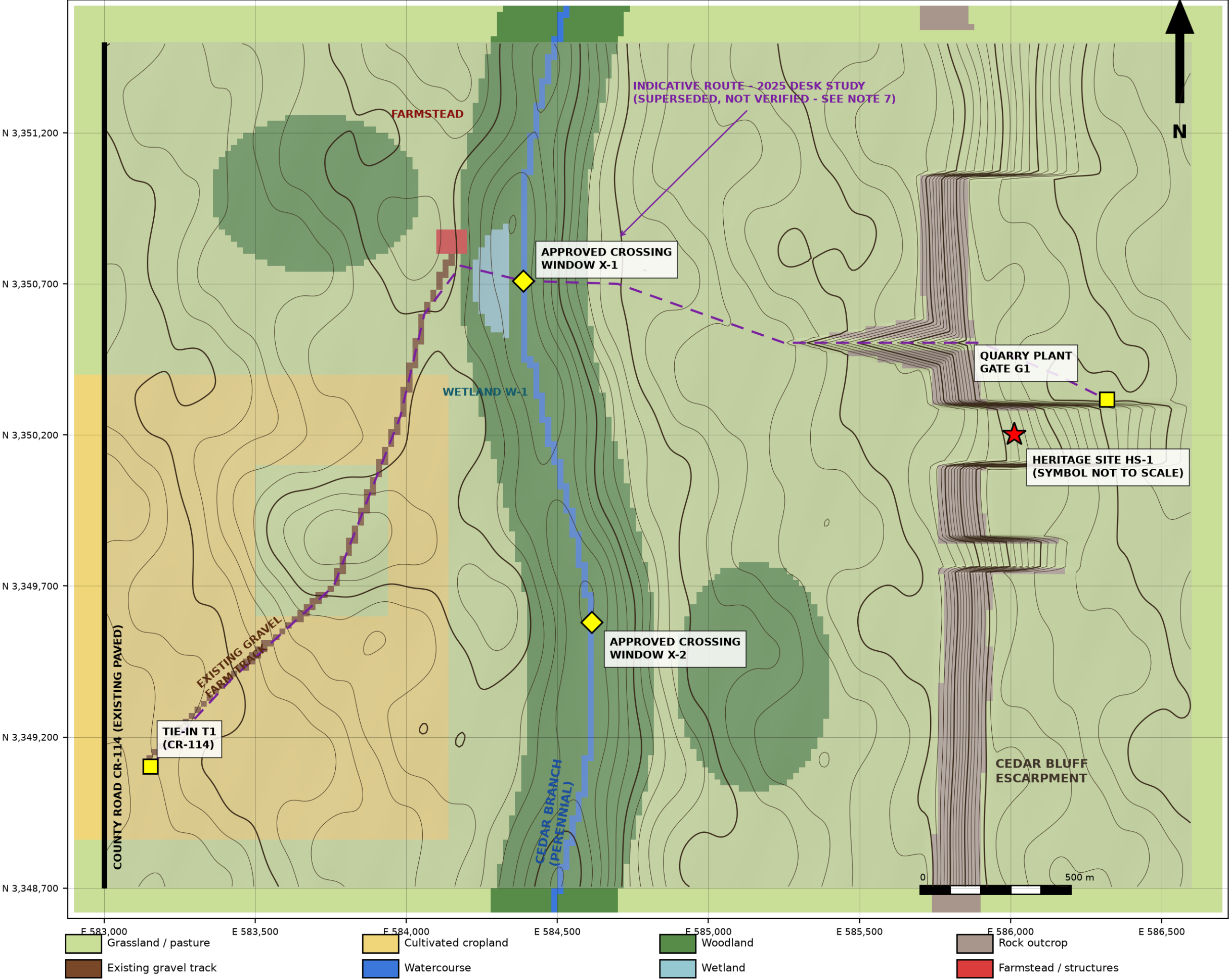


SITE CONSTRAINTS PLAN - PRESENTATION ONLY, DO NOT SCALE (SEE C-002 NOTE 1)



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PROJECT
Quarry Haul Road
Route Optioneering Study

DRAWING
**SITE CONSTRAINTS
PLAN**

REVISIONS

A 2026-08-03 Issued for route study

B 2026-09-14 Woodland clearing rate revised (C-002 Table 2)

C 2026-09-24 Plant gate G1 relocated; haul truck criteria and typical section added (Sheet C-003)

Rev C supersedes Rev B in full.

Horizontal: WGS 84 / UTM zone 14N (EPSG:32614), metres
Vertical: NAVD 88, US survey feet

NOT FOR CONSTRUCTION

SHEET
C-001

REV
C

Drawn: R.O. Checked: J.M. Date: 2026-09-24

Contours from CB-DEM-10m, 10 ft interval (50 ft index), NAVD 88 US survey feet. Land-cover colours as C-002 Table 2.

DATA REGISTER, UNIT RATES AND ROUTING CONSTRAINTS

TABLE 1 - RASTER DATA REGISTER (issued with this drawing)

Item	CB-DEM-10m_heightmap_uint16.png	CB-LC-20m_landcover.png
Content	Bare-earth terrain model (LiDAR 2026-03)	Land-cover classification (field verified 2026-07)
Raster size	360 columns x 280 rows, 1 band	190 columns x 150 rows, 3 bands
Pixel type	16-bit unsigned integer (DN)	8-bit RGB, classes per Table 2
CRS	WGS 84 / UTM zone 14N (EPSG:32614)	WGS 84 / UTM zone 14N (EPSG:32614)
UL corner of UL pixel	E 583 000.00 m N 3 351 500.00 m	E 582 900.00 m N 3 351 620.00 m
Pixel size	10.00 m x 10.00 m (square)	20.00 m x 20.00 m (square)
Row / column order	Row 1 = north edge, column 1 = west edge	Row 1 = north edge, column 1 = west edge
Value	Elevation = 1150.00 + 0.01 x DN (US survey feet, NAVD 88)	Exact RGB triplet = class (no anti-aliasing)

TABLE 2 - LAND-COVER CLASSES AND BASE UNIT RATES (REV B, UNCHANGED AT REV C)

Code	Class	RGB	Base unit rate c (USD per metre of road)
1	Grassland / pasture	(201, 222, 150)	420
2	Cultivated cropland	(240, 214, 120)	480
3	Woodland	(86, 140, 72)	940 (Rev B; Rev A value 690 withdrawn)
4	Rock outcrop	(168, 150, 140)	1,150
5	Existing gravel track	(120, 72, 40)	160
6	Watercourse	(60, 120, 220)	PROHIBITED (Note 3)
7	Wetland	(150, 200, 210)	PROHIBITED
8	Farmstead / structures	(220, 60, 60)	PROHIBITED
-	Culvert at approved crossing window	-	3,800 (Note 3)

TABLE 3 - CONTROL POINTS (WGS 84 geographic, decimal degrees)

Point	Latitude	Longitude
T1 Tie-in to CR-114 (route start)	30.2707418	-98.1355013
G1 Quarry plant gate (route end)	30.2814772	-98.1024981
X-1 Approved crossing window centre	30.2851604	-98.1225516
X-2 Approved crossing window centre	30.2749484	-98.1202827
HS-1 Heritage site (recorded point)	30.2804614	-98.1056944

NOTES

1. The two rasters in Table 1 govern. Sheet C-001 is a presentation plot of them and is not to be scaled or digitised.
2. Route start and end are the DEM cells containing control points T1 and G1.
3. Watercourse cells may be traversed only at approved crossing windows X-1 and X-2: a watercourse cell is passable where its DEM cell centre lies within 30.0 m of a window centre, and is then costed at the culvert rate. All other watercourse cells are prohibited.
4. Wetland W-1: wetland cells are prohibited and no part of the road may lie within 30.0 m of the mapped wetland boundary.
5. Heritage site HS-1: exclusion zone of radius 150.0 m about the recorded point (Texas Historical Commission consultation, 2026).
6. Maximum grade 10.0 percent for loaded 40 t haul trucks, in either direction of travel.
7. The indicative route on C-001 is from a 2025 desk study that pre-dates the LiDAR, the land-cover survey and Notes 3 to 5.
8. Haul truck operating criteria and the road typical section are on Sheet C-003. Where C-003 and these notes differ, C-003 governs.

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DATA REGISTER,

UNIT RATES AND

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NOT FOR CONSTRUCTION

SHEET

REV

C-002

C

Drawn: R.O. Checked: J.M. Date: 2026-09-24

HAUL ROAD TYPICAL SECTION AND HAUL TRUCK OPERATING CRITERIA

TYPICAL SECTION A-A (TANGENT, CUT/FILL NOT SHOWN) SCALE 1:100 AT A3 - DO NOT SCALE FROM THIS PLOT
CL (route centreline = DEM cell centres, C-002 Note 2)

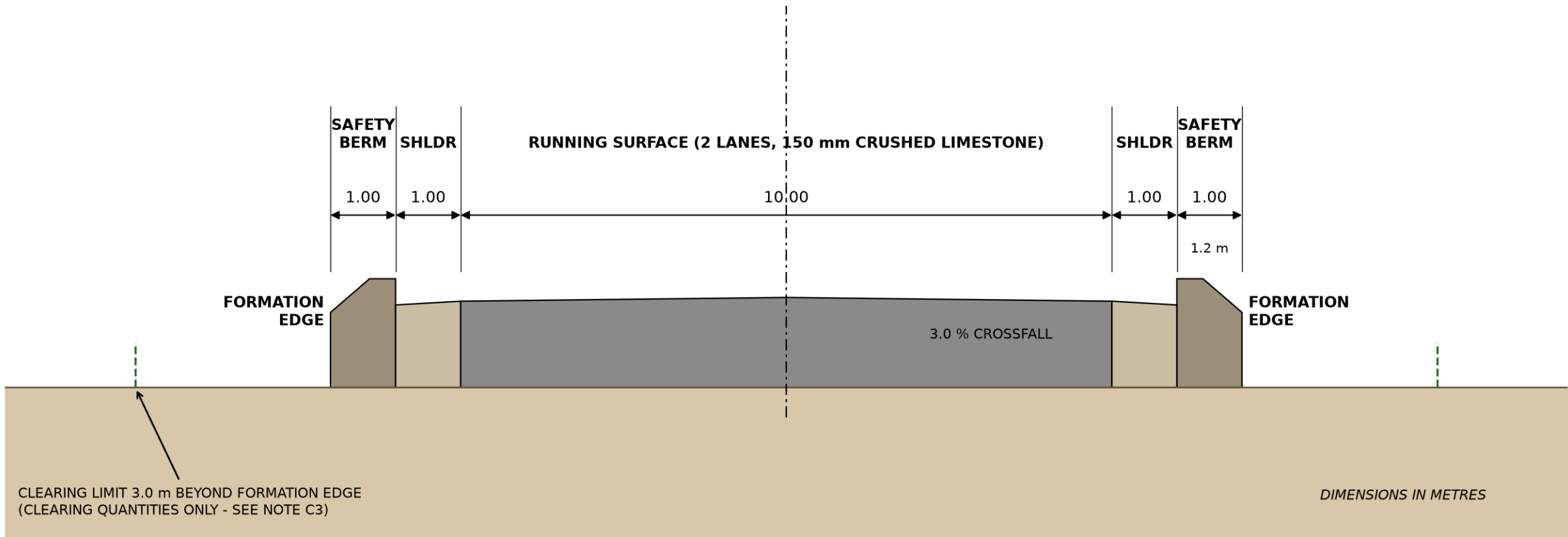


TABLE 4 - HAUL TRUCK OPERATING CRITERIA (40 t ARTICULATED HAUL TRUCK, LOADED)

Ref	Criterion	Value	Application to the route study
T1	Maximum grade, either direction	10.0 %	Every move; as C-002 Note 6
T2	Sustained steep grade	> 8.0 %	A move steeper than 8.0 % is a steep move
T3	Maximum continuous run of steep moves	60.0 m	Horizontal length; any move at 8.0 % or flatter ends the run
T4	Maximum change of direction between consecutive moves	45 deg	Measured between consecutive grid moves; the first move from the start cell is unrestricted
T5	Design speed (loaded)	40 km/h	Information only
T6	Rated payload / gross vehicle mass	40 t / 72 t	Information only

- NOTES**
- C1. 'The road' in C-002 Notes 4 and 5 means the full formation shown in Section A-A. Wetland setback and heritage exclusion are measured to the formation edge on either side of the centreline, along every straight move between cell centres.
- C2. The crossing-window test of C-002 Note 3 is applied to centreline cell centres only.
- C3. The clearing limit and batters are for quantities only and are not routing constraints.
- C4. Criteria T1 to T4 apply together. Unit rates (C-002 Table 2) and the grade factor are unchanged by this sheet.

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DRAWING
**TYPICAL SECTION
AND HAUL TRUCK
CRITERIA**

- REVISIONS**
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NOT FOR CONSTRUCTION

SHEET
C-003

REV
C

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